A Cauchy-Stable Partial Separation Reduction for the B_1^n Case

arXiv automatic math run `fa_banach_001`

July 3, 2026

Status

This packet gives a candidate partial result, likely valid pending expert review, for the Part 1 separation dimension-reduction challenge in Mendelson–Milman–Paouris [1]. It does not complete the authors' program: the separation loss obtained below is $O(k \log k)$, whereas the program needs a universal constant loss.

Source Challenge

The source paper is Shahar Mendelson, Emanuel Milman, and Grigoris Paouris, *Generalized Dual Sudakov Minoration via Dimension Reduction - A Program*, arXiv:1610.09287. In the conclusion, after noting that Part 2 is essentially settled up to the slicing problem, the authors write that the main remaining challenge is Part 1, and specifically mention $K = B_1^n$.

> μ is assumed 1-pure (e.g. super-Gaussian, sub-Gaussian or unconditional), then by Proposition 8.5, so will be all of its marginals $\nu \in \mathcal{M}_m$, for which we have a Weak Sudakov Minoration result by Theorem 9.1. In particular, up to the Slicing Problem, Part 2 is completely established.
>
> Consequently, it is clear that the main remaining challenge in completing The Program lies in establishing Part 1 of The Program. This is a significant challenge even
>
> 40
>
> for some specific convex bodies K besides ellipsoids, such as for $K = B_1^n$. To carry out this Separation Dimension-Reduction step, it seems that we would need to employ other measures on the Grassmannian $G_{n,m}$ besides the uniform Haar measure, upon which most of the (Euclidean) Asymptotic Geometric Analysis theory is built. In our opinion, this is a fascinating challenge, which we plan to explore in a future work.

Part 1' asks, informally, for a linear map to dimension $m \simeq k$ that keeps e^k many K-separated points separated while sending a positive amount of the original measure into a sufficiently massive target body. The authors suggest that for bodies such as B_1^n, one may need random maps distributed differently from Haar-random Euclidean projections.

Proof Intuition

For $K = B_1^n$, the natural non-Haar random map is a Cauchy matrix. The standard Cauchy law is 1-stable: every row applied to a vector x has scale $\|x\|_1$. Thus a difference with ℓ_1-norm > 1 has an exponentially small probability of landing in a fixed small ℓ_1^m-ball, so a union bound preserves e^k separations in $m \simeq k$ dimensions. On the other hand, vectors in B_1^n are mapped into an $O(m \log m) B_1^m$ ball with constant probability, giving the required massiveness after averaging over the measure. The $m \log m$ radius is the price paid for the heavy Cauchy tails, and is exactly why this remains only a partial result.

The Partial Result

Theorem 1 (Cauchy one-sided reduction for B_1^n)**.** *There are universal constants $a, A, c > 0$ such that the following holds. Let $k \geq 1$, $M \leq e^k$, and let $x_1, \ldots, x_M \in \mathbb{R}^n$ satisfy*

$$\|x_i - x_j\|_1 > 1 \qquad (i \neq j).$$

Let μ be a Borel probability measure on $\mathbb{R}^n$ with $\mu(B_1^n) \geq e^{-q}$, where $0 \leq q \leq k/2$. Put

$$m = \lceil k \rceil, \qquad L = A\, m \log(em)\, B_1^m.$$

Then there is a linear map $T : \mathbb{R}^n \to \mathbb{R}^m$ such that

$$\|Tx_i - Tx_j\|_1 > a \qquad (i \neq j)$$

and

$$T_* \mu(L) \geq c e^{-q}.$$

Equivalently, the images are $(1/R)L$-separated with

$$R \leq C\, m \log(em).$$

Lemma 2 (Two elementary Cauchy estimates)**.** *Let $G_1, \ldots, G_m$ be independent standard Cauchy random variables. There are universal constants $\eta, A > 0$ such that, for every $m \geq 1$,*

$$\mathbb{P}\left\{ \sum_{j=1}^m |G_j| \leq \eta \right\} \leq e^{-6m}$$

and

$$\mathbb{P}\left\{ \sum_{j=1}^m |G_j| \leq Am \log(em) \right\} \geq \frac{1}{2}.$$

Proof. For $0 < u \leq 1$, $\mathbb{P}\{|G_1| \leq u\} \leq Cu$. Choosing $\eta > 0$ so that $C\eta \leq e^{-6}$, the first estimate follows from the inclusion $\{\sum |G_j| \leq \eta\} \subseteq \{|G_j| \leq \eta \text{ for all } j\}$.

For the second estimate, use

$$\mathbb{P}\{|G_1| > u\} \leq \frac{C}{u}, \qquad \mathbb{E}\big[|G_1| \mathbf{1}_{\{|G_1| \leq u\}}\big] \leq C \log(eu).$$

With $R = Am\log(em)$,

$$\mathbb{P}\left\{\sum |G_j| > R\right\} \le \mathbb{P}\left\{\max_j |G_j| > R\right\} + \mathbb{P}\left\{\sum |G_j|\mathbf{1}_{\{|G_j|\le R\}} > R\right\}$$
$$\le \frac{Cm}{R} + \frac{Cm\log(eR)}{R}.$$

Taking A large enough makes the last quantity at most 1/2, uniformly in m. $\square$

Proof of Theorem 1. Let $T = (g_{jr})_{1\le j\le m, 1\le r\le n}$, where the g_{jr} are independent standard Cauchy random variables. The characteristic function identity $\mathbb{E}e^{itG} = e^{-|t|}$ implies the 1-stability relation

$$(Tx)_j \overset{d}{=} \|x\|_1\, G_j,$$

with independent G_j's as j varies.

If $h \notin B_1^n$, then $\|h\|_1 > 1$, and Lemma 2 gives

$$\mathbb{P}\{\|Th\|_1 \le \eta\} \le \mathbb{P}\left\{\sum_{j=1}^m |G_j| \le \eta\right\} \le e^{-6m}.$$

Therefore,

$$\mathbb{P}\{\exists i < j : \|T(x_i - x_j)\|_1 \le \eta\} \le M^2 e^{-6m} \le e^{2k-6m} \le e^{-4k}.$$

For $x \in B_1^n$, the same stability relation and the second part of Lemma 2 imply

$$\mathbb{P}\{Tx \in Am\log(em)B_1^m\} \ge \frac{1}{2}.$$

Consequently, for $L = Am\log(em)B_1^m$,

$$\mathbb{E}\big[T_*\mu(L)\big] = \int \mathbb{P}\{Tx \in L\}\, d\mu(x) \ge \frac{1}{2}\mu(B_1^n) \ge \frac{1}{2}e^{-q}.$$

Since $0 \le T_*\mu(L) \le 1$, this implies

$$\mathbb{P}\{T_*\mu(L) \ge \tfrac{1}{4}e^{-q}\} \ge \tfrac{1}{4}e^{-q}.$$

Because $q \le k/2$, the latter probability is larger than e^{-4k} after adjusting constants. Hence there is a realization of T for which both the separation event and the massiveness event occur. Taking $a = \eta$ and $c = 1/4$ proves the theorem. Finally,

$$\eta B_1^m = \frac{\eta}{Am\log(em)}L,$$

so the separation may be written as $(1/R)L$-separation with $R \le Cm\log(em)$. $\square$

Limitations and Novelty Check

The result is not the desired Part 1' for B_1^n, because R grows like $k\log k$. The source program needs R independent of k, n (or at least a substantially milder loss that can be absorbed in the bootstrap). The packet should therefore be read as evidence that 1-stable, non-Haar random maps are relevant to the B_1^n challenge, not as a completion of the program.

Novelty check performed on July 3, 2026: local run indexes were searched for arXiv:1610.09287, “Massive Separation Dimension Reduction”, B_1^n, Cauchy, and one-sided Johnson–Lindenstrauss terms; no duplicate packet was found. Web searches for “one-sided Johnson-Lindenstrauss Cauchy l1”, “Cauchy dimension reduction l1 separation”, and related stable-projection phrases did not surface a direct match. The result uses only elementary Cauchy stability and is not claimed to be absent from the broader algorithmic dimension reduction literature.

Human Review Recommendation

Reviewers should check whether the $O(k \log k)$ loss can be reduced by using a different target body, truncating the Cauchy map, or replacing the ℓ_1^m massiveness body by a Lorentz or weak-ℓ_1 body compatible with Part 2. As written, the proof is self-contained and the main risk is relevance rather than correctness.

References

[1] S. Mendelson, E. Milman, and G. Paouris, *Generalized Dual Sudakov Minoration via Dimension Reduction - A Program*, arXiv:1610.09287, 2016.